Notes: Nadauld & Sherlund (JFE, 2013)

The Impact of Securitization on the Expansion of Subprime Credit

Contents

1 Big picture

- **Question.** Did growth in private-label subprime mortgage securitization cause the expansion of subprime credit in the run-up to the financial crisis?
- **Main empirical challenge.** Securitization and subprime originations rise together, but this correlation alone cannot distinguish whether securitization caused credit expansion or merely responded to increased primary-market loan supply.
- **Core identification idea.** Compare securitization activity by the five large independent broker/dealer investment banks with securitization activity by other securitizing banks within the same ZIP code. The authors build an *Excess Demand* variable equal to growth in securitized loans by investment banks minus growth in securitized loans by non-investment banks.
- **Treatment group.** The five broker/dealer investment banks are Bear Stearns, Lehman Brothers, Merrill Lynch, Morgan Stanley, and Goldman Sachs.
- **Why these banks matter.** Their market share in subprime securitization rose sharply from 2003 to 2005. The paper argues that the increase reflects securitization-market forces tied to broker/dealer investment banks rather than ZIP-code-level borrower fundamentals.
- **Main result.** ZIP codes with higher investment-bank excess demand experienced more subprime originations, lower denial rates, and higher subsequent delinquency and default rates.
- **Magnitude.** A one-standard-deviation increase in excess demand is associated with roughly 2.5–6.5% higher subprime mortgage origination rates per housing unit and about 0.5% lower denial rates.
- **Consequences.** ZIP codes with higher excess demand experienced higher later delinquency and default rates, indicating that the credit expansion was not merely safe credit deepening.
- **Mechanism.** The paper finds evidence more consistent with reduced lender screening incentives than with a pure lower-cost-of-capital story: interest rates rely more heavily on hard information, dispersion in interest rates falls, and no/low documentation lending rises in high excess-demand ZIP codes.
- **Event validity.** The paper uses a 2004 Standard & Poor's change in treatment of assignee liability laws as an external shock to securitization costs. The shock lowered investment-bank securitization in affected states and supports the interpretation that the excess-demand variable captures secondary-market forces.
- **Takeaway.** Securitization did not simply passively absorb loans already produced by primary markets. The growth in private-label securitization by investment banks actively expanded subprime credit and weakened screening incentives.

2 Incremental contribution of the paper

2.1 Relative to the literature on the credit boom

- Mian and Sufi document that subprime ZIP codes experienced much faster mortgage credit growth than prime ZIP codes in the early 2000s.
- Nadauld and Sherlund move from documenting credit growth to identifying a specific causal channel: secondary-market demand from private-label securitization.
- The paper asks whether securitization was an active driver of credit expansion rather than a passive take-out mechanism for loans already originated for primary-market reasons.

Incremental contribution: The paper contributes a strategy for isolating secondary-market demand from primary-market fundamentals by comparing different securitizing-bank types within the same local mortgage markets.

2.2 Relative to the literature on securitization and screening

- Keys, Mukherjee, Seru, and Vig and related studies show that securitized loans can perform worse and that securitization may weaken screening around credit-score thresholds.
- Nadauld and Sherlund study whether the *increase in securitization demand itself* changed the quantity and composition of credit across ZIP codes.
- They combine quantity outcomes, denial-rate outcomes, default outcomes, interest-rate dispersion, and documentation status in one empirical framework.

Incremental contribution: The paper links securitization-induced credit expansion to a screening mechanism at the local-market level, rather than only comparing securitized and non-securitized loans ex post.

2.3 Relative to studies of investment banks and structured finance

- Much of the crisis literature emphasizes the role of broker/dealer investment banks in funding, structuring, and distributing mortgage-backed securities.
- This paper uses the cross-bank variation directly: the five large investment banks form a treatment group whose securitization activity rose sharply relative to other banks.
- The identification does not require knowing exactly why investment banks expanded; it requires that their expansion be tied to secondary-market forces rather than ZIP-code borrower demand.

Incremental contribution: The paper operationalizes the investment-bank role in the credit boom by turning it into a measurable ZIP-level excess-demand variable.

2.4 Relative to simple correlation studies of securitization

- A regression of credit extension on the percent of loans sold to the secondary market is endogenous because both variables respond to local borrower demand, expected house price appreciation, and lender risk preferences.
- The authors use matched LoanPerformance–ABSNet data to identify which securitizing banks purchased loans from each ZIP code.
- They construct a difference-in-growth measure that compares investment-bank securitization growth with noninvestment-bank securitization growth in the same market.

Incremental contribution: The incremental methodological contribution is the construction of a secondary-market demand shifter designed to remove ZIP-level primary-market confounds.

2.5 What the paper adds to policy debates

- It provides evidence that securitization can expand credit supply through originator incentives, not only through funding liquidity.
- It shows that secondary-market demand can reduce screening discipline and increase risk in the underlying loans.
- It suggests that regulation of securitization markets and the incentives of security underwriters can matter for household credit allocation.

Incremental contribution: The paper's policy contribution is to connect the structure and incentives of private-label securitization to credit access, loan quality, and default risk in the primary mortgage market.

3 Data and measurement

3.1 Data sources

- **LoanPerformance.** Provides individual subprime loan information, including loan characteristics, origination ZIP code, performance, and default/delinquency outcomes.
- **ABSNet.** Provides information on subprime residential mortgage-backed securitization deals, including deal underwriters and deal-level collateral information.
- **HMDA.** Provides mortgage application and origination data and permits a higher-priced loan definition of subprime credit.
- **HUD subprime lender list.** Provides an alternative definition of subprime lending based on whether loans are originated by lenders classified as subprime specialists.
- **Census and local controls.** Supply ZIP/MSA-level demographic and economic controls such as average income, credit quality, homeownership, permits, unemployment, and house-price appreciation.

Interpretation: The distinctive data contribution is the hand match between individual loans and ABSNet securitization deals. This allows the authors to identify not just whether a loan is securitized, but which type of securitizing bank is associated with securitized loans from a given ZIP code.

3.2 Securitization treatment and control groups

- The treated securitizers are the five independent broker/dealer investment banks: Bear Stearns, Lehman Brothers, Merrill Lynch, Morgan Stanley, and Goldman Sachs.
- The control group includes all other securitizing banks.
- The paper's key cross-sectional variation comes from how much the treated banks increased securitization relative to the control banks within a ZIP code.

Interpretation: Comparing treated and control securitizers in the same ZIP code helps remove the local borrower-quality and local housing-market variables that would affect all securitizers active in that ZIP code.

3.3 Excess Demand

The main secondary-market demand measure is:

$$ExcessDemand_{i,t} = \left(\frac{SecLoans^{Treated}_{i,T}}{SecLoans^{Treated}_{i,T-t}}\right) - \left(\frac{SecLoans^{Control}_{i,T}}{SecLoans^{Control}_{i,T-t}}\right), \tag{1}$$

where i indexes ZIP codes, T is the end year, and t is the lag window.

- The authors use 2003–2004 and 2003–2005 versions.
- A higher value means investment-bank securitization grew more rapidly than noninvestment-bank securitization in the same ZIP code.
- The variable is interpreted as a secondary-market demand shock.

Interpretation: The variable is not simply the level of securitization. It is a relative growth measure that attempts to isolate the investment-bank-driven component of securitization demand.

3.4 Credit extension outcomes

The main credit extension outcomes are:

- log of higher-priced subprime loans per housing unit under HMDA;
- log of HUD subprime loans per housing unit;
- ZIP-code mortgage denial rates;
- percent of loans sold to the secondary market, used as a conventional benchmark measure.

Interpretation: The HMDA and HUD definitions provide two different ways to measure subprime credit. The denial-rate measure captures whether credit access widened on the extensive margin.

3.5 Loan performance outcomes

The paper studies adverse credit outcomes using LoanPerformance data:

- delinquency: loans 90 days or more delinquent at any point during their existence;
- default: loans that are in foreclosure for at least two consecutive months;
- outcomes are measured for 2003, 2004, and 2005 origination vintages as of December 2010.

Interpretation: The vintage comparison asks whether the securitization-driven credit expansion was associated with lower loan quality over time.

3.6 Screening and interest-rate measures

To distinguish reduced screening from lower-cost-of-capital channels, the paper studies:

- the standard deviation of adjustable mortgage interest rates within ZIP codes;
- average ZIP-code interest rates;
- the explanatory power of hard-information variables such as FICO, loan-to-value, and debt-to-income;
- the share of low or no documentation loans.

Interpretation: If securitization weakens soft-information screening, lenders should rely more on hard variables and less on costly soft information. This should reduce unexplained interest-rate dispersion and increase low-documentation lending.

4 Empirical specifications and models

4.1 Baseline credit extension specification

The core cross-sectional regression is:

$$Y_{i,t} = \alpha + \beta ExcessDemand_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}, \tag{2}$$

where $Y_{i,t}$ is a ZIP-code credit outcome and $X_{i,t}$ contains demographic and mortgage-market controls.

- In Tables 3 and 4, Y is log subprime loans per housing unit.
- In Table 5, Y is the ZIP-code mortgage denial rate.
- Standard errors are clustered at the state level.

Interpretation: The coefficient β measures whether ZIP codes with more investment-bank excess demand experienced greater credit expansion relative to similar ZIP codes.

4.2 Benchmark comparison with percent sold to secondary market

The paper also estimates:

$$Y_{i,t} = \alpha + \delta PercentSold_{i,t} + \Gamma X_{i,t} + \varepsilon_{i,t}. \tag{3}$$

Interpretation: The percent-sold variable is a conventional securitization proxy, but it is more endogenous than Excess Demand. Comparing the two measures shows whether the new instrument-like variable generates similar economic patterns without relying on the endogenous percent-sold measure.

4.3 Adverse outcome regressions

For delinquencies and defaults, the paper estimates changes across vintages:

$$\Delta DefaultOrDelinquency_i = \alpha + \beta ExcessDemand_i + \Gamma X_i + \varepsilon_i. \tag{4}$$

Interpretation: The regression asks whether places that received larger securitization-driven credit expansion also experienced worse ex post loan performance.

4.4 Lax screening versus lower cost of capital

The paper contrasts two mechanisms:

- **Lower cost of capital.** Securitization expands credit because lenders can fund loans more cheaply. This should allow lower-quality borrowers to receive loans, but it does not necessarily imply less screening.
- **Lax screening.** Securitization weakens lenders' incentives to collect and price soft information. Lenders rely more on observable hard information and originate more low-documentation loans.

A typical interest-rate dispersion regression is:

$$\Delta SD(InterestRate_i) = \alpha + \beta ExcessDemand_i + \Gamma X_i + \varepsilon_i. \tag{5}$$

A typical interest-rate level regression is:

$$InterestRate_i = \alpha + \Gamma HardInfo_i + \varepsilon_i, \tag{6}$$

with separate comparisons for positive versus negative excess-demand ZIP codes.

Interpretation: A lax-screening channel predicts that hard variables explain more of pricing in high-securitization markets, because soft information is used less intensively.

4.5 Assignee liability event test

In May 2004, S&P changed how it treated securitization deals containing loans from states with vague or uncertain assignee liability laws. The paper uses Massachusetts as a key affected state.

The event specification uses interactions of the form:

$$Y_{i,b,t} = \alpha + \beta_1 InvBank_b \times MA_i + \beta_2 MA_i \times Event_t + \beta_3 InvBank_b \times Event_t + \beta_4 InvBank_b \times MA_i \times Event_t + \Gamma X_{i,t} + \varepsilon_{i,b,t}. \tag{7}$$

Interpretation: If the S&P change increased the cost of securitizing affected loans, investment banks should reduce securitization in Massachusetts after the event. This validates that the secondary-market demand measure responds to securitization-market costs.

5 Identification and empirical design

5.1 The omitted-variable problem

A simple regression of credit extension on securitization is endogenous because:

- stronger expected house price appreciation raises both lending and securitization demand;
- better borrower characteristics raise both origination and secondary-market saleability;
- local lender risk preferences or marketing intensity may affect both loan supply and securitization;
- loan originators may create loans specifically because they know they can sell them.

Interpretation: The paper's identification strategy is designed to avoid interpreting any increase in securitization as causal unless it is tied to differences in securitization activity that are plausibly separate from local borrower demand.

5.2 Why within-ZIP bank-type comparisons help

- A ZIP code's macroeconomic and demographic characteristics should affect all securitizing banks active in that ZIP code.
- If investment banks increase securitization more than other banks within the same ZIP code, the difference is less likely to be driven by ZIP-level borrower fundamentals.
- The difference is instead interpreted as an investment-bank-specific securitization demand shock.

Interpretation: The identification rests on comparing two types of securitizers facing the same local mortgage environment.

5.3 Why the 2003–2005 period matters

- The subprime mortgage boom and securitization boom were both concentrated in 2003–2005.
- The market share of broker/dealer investment banks rose sharply during this period.
- The paper uses growth in securitization activity over 2003–2004 and 2003–2005 to capture this burst.

Interpretation: The identification is strongest when the treated banks show a distinctive jump relative to other securitizers.

5.4 Event validity

- The S&P assignee liability change affected the cost of securitization, not borrower demand directly.
- Investment banks reduce securitization activity more in affected Massachusetts ZIP codes after the event.
- The same shock translates into fewer originations and higher denial rates in affected areas.

Interpretation: The event analysis is a validation exercise: it shows that changing securitization costs affects the excess-demand channel and primary-market outcomes.

5.5 Threats and limitations

- Exact reasons for investment banks' expansion are not pinned down.
- The Excess Demand measure may still capture some unobserved bank-specific strategy correlated with local markets.
- ZIP-level regressions cannot observe every borrower-level soft-information decision.
- Some outcomes depend on the accuracy of matching LoanPerformance loans to ABSNet securitization deals.

Interpretation: The paper's strength is not a randomized assignment of securitization demand, but a layered strategy: within-ZIP treated/control securitizers, multiple outcome definitions, screening tests, and an external securitization-cost event.

6 Tables and figures

6.1 Figures 1–3 and Tables 1–2: building the excess-demand measure

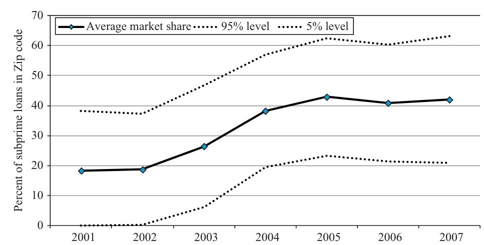
Read:

- Figure 1 shows that subprime securitization underwriting activity increased sharply and was concentrated among a small set of underwriters.
- Figure 2 shows that the five broker/dealer investment banks' market share rose from roughly the low 30% range in 2003 to a peak near 48% in 2005.
- Figure 3 shows that all five treated investment banks exhibit positive excess demand mainly in 2004–2005.
- Table 1 summarizes primary-market subprime activity and secondary-market activity at the ZIP-code level.
- Table 2 shows that bank-ZIP-year observations for investment banks and noninvestment banks are comparable on average, but investment banks' securitization activity rises strongly during the boom.

Economic message: The excess-demand measure is motivated by a real institutional shift: broker/dealer investment banks expand subprime securitization faster than other securitizers in the same markets.

Underwriter	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	Total
Lehman Brothers	0	0	4	3	3	8	16	20	31	31	14	130
Bear Stearns & Co. Inc	0	0	3	1	1	1	6	23	34	27	16	112
Greenwich Capital	0	0	1	5	11	9	13	18	20	24	9	110
Morgan Stanley	1	1	0	1	0	4	12	29	29	20	13	110
Credit Suisse	0	1	1	0	8	10	13	23	25	13	6	100
Merrill Lynch	0	0	2	0	1	0	4	12	31	34	9	93
Deutsche Bank Securities Inc.	0	0	1	1	2	7	13	15	20	24	8	91
Goldman Sachs	0	0	0	0	0	3	5	17	20	22	9	76
Bank of America	0	0	0	2	3	8	14	18	11	6	5	67
Citigroup Global Markets Inc.	0	0	0	0	0	2	6	9	16	17	14	64
JP Morgan	0	0	0	0	5	7	7	4	8	21	8	60
UBS	0	0	0	0	0	1	8	13	15	20	3	60
Barclays	0	0	0	0	0	0	0	8	15	19	8	50
RBS Greenwich	0	0	0	0	0	1	3	10	10	6	8	38
Countrywide Securities Corp.	0	0	0	0	0	4	8	14	5	0	0	31
HSBC	0	0	0	0	0	0	0	0	4	13	6	23
Residential Funding Corp.	0	0	0	0	0	0	0	0	5	8	1	14
Washington Mutual	0	0	0	0	0	0	0	0	0	9	0	9
Prudential Securities	0	2	2	2	0	0	0	0	0	0	0	6
Unknown	4	1	0	0	0	0	0	0	1	0	0	6
Nomura	0	0	0	0	0	0	0	0	0	5	0	5
Salomon Smith Barney	0	0	0	0	0	5	0	0	0	0	0	5
GMAC RFC	0	0	0	0	0	0	1	0	3	0	0	4
BNP Paribas	0	0	0	0	0	0	0	0	1	2	0	3
GMACM Mortgage Corp.	0	0	0	0	0	0	3	0	0	0	0	3
Banc One	0	0	0	0	0	0	1	0	0	0	0	1
Bank of New York	0	0	0	0	0	0	0	1	0	0	0	1
Blaylock & Company	0	0	0	0	0	0	1	0	0	0	0	1
Carrington	0	0	0	0	0	0	0	0	0	1	0	1
Chase	0	0	0	1	0	0	0	0	0	0	0	1
Donaldson Lufkin & Jenrette	0	1	0	0	0	0	0	0	0	0	0	1
Residential Asset Securities Corp.	0	0	0	0	0	0	0	0	0	1	0	1
SG Americas Securities LLC	0	0	0	0	0	0	0	0	0	1	0	1
Saxon Asset Securities Company	0	0	0	0	0	0	0	0	0	0	1	1
Utendahl Capital Partners	0	0	0	0	0	0	0	1	0	0	0	1
Total	5	6	14	16	34	70	134	235	304	324	138	1280
Five broker/dealer investment bank total	1	1	9	5	5	16	43	101	145	134	61	521
Five broker/dealer investment bank share of total	20.0%	16.7%	64.3%	31.3%	14.7%	22.9%	32.1%	43.0%	47.7%	41.4%	44.2%	40.7%
All other banks total	4	5	5	11	29	54	91	134	159	190	77	759
All other banks share of total	80.0%	83.3%	35.7%	68.8%	85.3%	77.1%	67.9%	57.0%	52.3%	58.6%	55.8%	59.3%

(a) Figure 1: subprime securitization deals by underwriter.



vestment banks relative market share plotted through time. The solid line represents the average share of all securit
ed by the five independent broker/dealer investment banks (referred to as the treatment sample of banks in the te

(b) Figure 2: investment-bank market share.

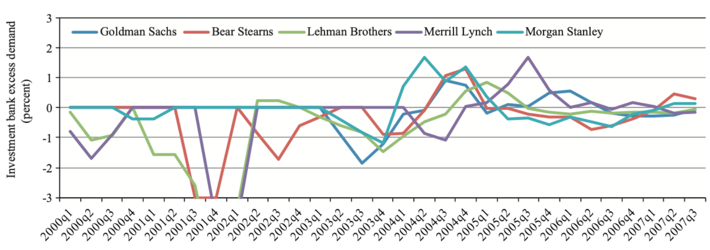


Fig. 3. Broker/dealer investment bank excess demand. This figure plots investment bank-specific excess demand through time. *Excess Demand* is calculated as the difference between the growth in the number of securitized loans of broker/dealer investment banks compared with the growth in the number of securitized loans by all other banks. *Excess Demand* is measured at the ZIP code level and is calculated in this chart as

$$\textit{Excess Demand}_{\textit{investment bank } i,t} = \left(\frac{\#\textit{Sec. Loans,Investment Bank}_{i,t}}{\#\textit{Sec. Loans,Investment Bank}_{i,t-1}} - \frac{\#\textit{Sec. Loans,All Other Banks}_{i,t}}{\#\textit{Sec. Loans,All Other Banks}_{i,t-1}}\right).$$

Figure 3: investment-bank excess demand over time.

6.2 Tables 3–5: credit expansion and denial rates

Read:

Table 1

Summary statistics of secondary market activity at ZIP code level.

Panel A reports summary statistics on mortgage origination activity for the higher-priced definition of subprime loans using the Home Mortgage Disclosure Act (HMDA) data set. Panel B reports summary statistics on mortgage origination activity for the Department of Housing and Urban Development subprime lender definition of subprime loans in the HMDA data set. Panel C reports summary statistics on secondary market activity for broker/dealer investment banks and noninvestment banks. Growth rates from 2003 to 2004, 2004 to 2005, and 2003 to 2005 are computed as the percent change in the total number of subprime loans securitized in a given ZIP code over the respective time periods.

	Year	Number of ZIP codes	10%	Median	Mean	90%	Standard deviation
Panel A: HMDA higher-priced summary statistics							
Subprime	2004	14,995	0.010	0.022	0.034	0.058	0.057
Loans per housing unit	2005	15,139	0.017	0.037	0.058	0.116	0.076
Percent of loans sold	2004	15,067	0.417	0.654	0.621	0.773	0.145
	2005	15,259	0.500	0.720	0.685	0.814	0.129
Panel B: HUD subprime lender list summary statistics							
Subprime	2003	15,353	0.001	0.013	0.019	0.034	0.037
Loans per housing unit	2004	14,862	0.006	0.016	0.025	0.045	0.039
	2005	13,552	0.006	0.016	0.026	0.049	0.042
	2003	12,574	0.435	0.700	0.644	0.810	0.202
Percent of loans sold	2004	16,002	0.532	0.720	0.694	0.813	0.123
	2005	18,361	0.550	0.755	0.724	0.852	0.138
Panel C: Secondary market summary statistics							
Inv. bank % increase	2003–2004	17,132	0.000	0.887	0.854	3.931	0.649
	2004–2005	20,640	−0.511	0.205	0.207	0.847	0.574
	2003–2005	17,154	0.000	1.098	1.073	1.945	0.716
Non-inv. bank % increase	2003–2004	23,343	−0.405	0.342	0.328	1.061	0.584
	2004–2005	23,951	−0.667	0.000	0.063	0.693	0.550
	2003–2005	22,926	−0.373	0.405	0.396	1.098	0.626
Inv. bank % minus non-inv. bank %	2003–2004	16,031	−0.405	0.511	0.502	1.406	0.763
	2004–2005	18,941	−0.693	0.193	0.158	0.981	0.742
	2003–2005	16,107	−0.288	0.707	0.674	1.579	0.775

Table 1: summary statistics at the ZIP-code level.

Table 2

Summary statistics on bank-ZIP code-year panel data.

Our panel data set of bank-ZIP code-years runs from 1997 to 2007. Matching the LoanPerformance and ABSNet data allows for the identification of loans in each ZIP code in each year that were securitized by the five broker/dealer investment banks and noninvestment banks. This table reports the average attributes of the bank-ZIP code-year data through time. House price appreciation (H.P.A.) is measured using ZIP code level indexes when available and MSA and state-level indexes when ZIP code indexes are unavailable. Data on FICO scores and debt-to-income ratios are missing early in the sample period.

	Broker/dealer investment banks					Noninvestment banks				
Year	Number of bank-ZIP code-years	LTV	HPAt-1	FICO	DTI	Number of bank-ZIP code-years	LTV	HPAt-1	FICO	DTI
1997	1,716	72.46	3.09	–	–	5,345	74.69	2.99	–	–
1998	5,092	75.97	5.09	–	–	9,473	73.75	4.45	374	–
1999	24,884	76.60	5.87	440	25.07	16,122	76.49	6.46	459	31.33
2000	13,408	75.00	7.25	524	19.15	28,298	76.99	7.44	519	24.60
2001	19,312	78.51	7.64	582	17.99	53,848	79.01	7.17	547	28.87
2002	26,047	78.92	7.09	571	28.35	91,400	78.86	7.26	595	21.64
2003	48,830	79.92	8.28	607	28.24	129,681	80.75	8.31	608	26.17
2004	74,950	83.10	9.82	605	31.39	150,535	82.52	10.03	610	26.67
2005	80,511	85.49	11.43	617	32.49	168,461	84.27	11.97	613	25.23
2006	78,076	86.32	9.13	616	36.26	176,853	86.32	9.68	613	33.07
2007	55,653	85.76	4.13	617	35.86	97,113	84.68	4.00	613	35.63

Table 2: bank-ZIP-year summary statistics.

- Table 3 uses the HMDA higher-priced definition of subprime loans. Excess demand is positively related to subprime originations in both 2004 and 2005.
- Table 4 repeats the test using the HUD subprime lender definition. The positive relation remains.
- Table 5 shows that higher excess demand is associated with lower ZIP-code denial rates, especially over 2003–2005.
- Conventional percent-sold measures have similar signs but are more endogenous than the excess-demand variable.

Economic message: The tables provide the core credit-supply evidence: secondary-market demand from investment banks increased the quantity of subprime credit and reduced rejection of loan applications.

Table 3

Secondary market demand and access to credit: higher-priced definition of subprime.

We estimate cross-sectional regressions on data from the years 2004 and 2005 separately. The regressions measure the impact of demand from the secondary mortgage market on the extension of credit in the primary mortgage market. Access to credit in the primary mortgage market is measured as the natural log of the number of originated subprime loans as a fraction of total housing units in a given ZIP code. Subprime loan originations are measured using the Home Mortgage Disclosure Act higher-priced definition. See Appendix for details. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. We discuss the construction and aggregation of the control variables to the ZIP code level in the Appendix. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1, 5, and 10% level, respectively.

	Dependent variable: Log of higher-priced subprime loans per housing unit			
	2004 (1)	2005 (2)	2004 (3)	2005 (4)
Inv. bank % increase minus non-inv. bank % increase	0.040***			
(2003–2004)	(2.803)			
Inv. bank % increase minus non-inv. bank % increase		0.087***		
(2003–2005)		(4.711)		
% Subprime Sold to Secondary Market			0.604***	1.521***
			(2.789)	(6.195)
	0.021***	0.025***	0.015***	0.017***
	(4.028)	(6.611)	(2.850)	(4.470)
MSA avg. income—Quartile 1	0.250	0.565	−0.131	0.094
	(0.754)	(1.574)	(-0.403)	(0.297)
MSA avg. income—Quartile 2	0.303	0.656*	−0.126	−0.062
	(0.909)	(1.861)	(−0.419)	(−0.219)
MSA avg. income—Quartile 3	0.217	0.629*	−0.302	−0.094
	(0.601)	(1.782)	(−0.982)	(−0.415)
MSA avg. income—Quartile 4	0.192	0.479	−0.245	−0.007
	(0.505)	(1.098)	(−0.718)	(−0.0221)
Percent low-bucket credit	2.265***	1.662***	2.124***	1.700***
	(5.805)	(3.860)	(4.939)	(3.772)
Percent medium-bucket credit	6.309***	6.129***	5.475***	5.911***
	(7.913)	(6.476)	(7.875)	(7.989)
Home ownership rate	0.004**	0.002	0.004	0.002
	(2.039)	(0.938)	(1.585)	(0.898)
Housing permits one year lag	0.125***	0.124***	0.124***	0.118***
	(5.835)	(7.036)	(5.518)	(6.311)
Unemployment rate	0.006	0.006	−0.004	0.004
	(0.381)	(0.244)	(−0.260)	(0.216)
Constant	−5.931***	−5.450***	−5.683***	−5.899***
	(−13.37)	(−11.49)	(−11.37)	(−11.27)
Number of observations	11,046	11,058	15,028	15,173
Adjusted *R*-squared	0.315	0.300	0.245	0.271
Cluster by state	Yes	Yes	Yes	Yes

Table 4

Secondary market demand and access to credit: Department of Housing and Urban Development (HUD) measure of subprime.

We estimate cross-sectional regressions on data from the years 2004 and 2005 separately. The regressions measure the impact of demand from the secondary mortgage market on the extension of credit in the primary mortgage market. Access to credit in the primary mortgage market is measured as the natural log of the number of originated subprime loans as a fraction of total housing units in a given ZIP code. Subprime loan originations are measured using the HUD subprime lender list definition. See Appendix for details. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. We discuss the construction and aggregation of the control variables to the ZIP code level in the Appendix. Standard errors are clustered at the state level. ***, ** and * indicate significance at the 1%, 5%, and 10% level, respectively.

	Dependent variable: Log of HUD subprime loans per housing unit			
	2004 (1)	2005 (2)	2004 (3)	2005 (4)
Inv. bank % increase minus non-inv. bank % increase	0.033			
(2003–2004)	(1.198)			
Inv. bank % increase minus non-inv. bank % increase		0.087***		
(2003–2005)		(4.019)		
% Subprime Sold to Secondary Market			0.795	1.604***
			(1.529)	(3.619)
HPA growth lag one year	0.042***	0.036***	0.039***	0.033***
	(5.232)	(7.034)	(4.458)	(5.865)
MSA avg. income—Quartile 1	1.056**	1.222**	0.826*	0.850*
	(2.574)	(2.649)	(1.944)	(1.951)
MSA Avg. Income—Quartile 2	1.298***	1.425***	1.023**	0.873**
	(3.603)	(3.478)	(2.648)	(2.065)
MSA avg. income—Quartile 3	1.274***	1.371***	0.847**	0.819**
	(3.940)	(3.441)	(2.091)	(2.396)
MSA avg. income—Quartile 4	1.386***	1.428**	0.938*	0.948*
	(2.725)	(2.546)	(1.987)	(1.957)
Percent low-bucket credit	1.554***	1.968***	1.620***	2.000***
	(4.234)	(4.666)	(4.240)	(4.662)
Percent medium-bucket credit	6.119***	5.288***	4.410***	3.839***
	(5.646)	(3.706)	(5.129)	(3.548)
Home ownership rate	0.023***	0.024***	0.023***	0.024***
	(9.299)	(7.104)	(9.154)	(7.559)
Housing permits one year lag	0.143***	0.110***	0.156***	0.124***
	(5.323)	(5.371)	(5.676)	(4.951)
Unemployment rate	0.036*	0.012	0.033*	0.015
	(1.892)	(0.512)	(1.942)	(0.676)
Constant	−8.401***	−8.417***	−8.517***	−9.085***
	(−17.89)	(−14.92)	(−17.25)	(−15.47)
Number of observations	9,791	9,000	14,135	13,058
Adjusted *R*-squared	0.298	0.306	0.237	0.265
Cluster by state	Yes	Yes	Yes	Yes

6.3 Table 6: consequences of credit expansion

Read:

- Table 6 studies changes in ZIP-code delinquency and default rates for 2004 and 2005 loan vintages relative to 2003.

- The 2003–2005 excess-demand coefficient is positive and significant for delinquency and default outcomes.
- ZIP codes with higher initial house price appreciation also show increases in adverse outcomes.

Economic message: Securitization-driven credit expansion was associated with worse loan performance, so the increase in credit was not simply a benign expansion to safe borrowers.

Table 5

Secondary market demand and access to credit: ZIP code denial rate.

We estimate cross-sectional regressions on data from the years 2004 and 2005 separately. The regressions measure the impact of demand from the secondary mortgage market on the extension of credit in the primary mortgage market. Access to credit in the primary mortgage market is measured as the fraction of loan applications in a ZIP code that are denied. Subprime loan originations are measured using the Department of Housing and Urban Development subprime lender list definition. See Appendix for details. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. We discuss the construction and aggregation of the control variables to the ZIP code level in the Appendix. Standard errors are clustered at the state level. ***,**, and * indicate significance at the 1%, 5%, and 10% level, respectively. Robust *t*-statistics in parentheses.

	Dependent variable: ZIP Code denial rate			
	2004 (1)	2005 (2)	2004 (3)	2005 (4)
Inv. Bank % increase minus non-inv. Bank % increase (2003–2004)	−0.001			
	(−1.019)			
Inv. Bank % Increase Minus Non-Inv. Bank % Increase (2003–2005)		−0.008***		
		(−4.993)		
% Subprime sold to secondary market			−0.074***	−0.117***
			(−5.177)	(−6.456)
HPA growth lag one year	−0.001***	−0.001***	−0.001**	−0.001***
	(−2.788)	(−4.578)	(−2.143)	(−2.910)
MSA Avg. Income—Quartile 1	−0.054*	−0.053	−0.053*	−0.026
	(−1.967)	(−1.546)	(−1.924)	(−0.746)
MSA Avg. Income—Quartile 2	−0.051*	−0.034	−0.046*	−0.007
	(−1.742)	(−1.121)	(−1.828)	(−0.263)
MSA Avg. Income—Quartile 3	−0.085***	−0.078***	−0.075***	−0.048**
	(−3.209)	(−3.057)	(−3.377)	(−2.223)
MSA Avg. Income—Quartile 4	−0.068**	−0.062**	−0.055**	−0.025
	(−2.653)	(−2.337)	(−2.362)	(−0.910)
Percent low-bucket credit	0.465***	0.457***	0.424***	0.399***
	(22.93)	(24.83)	(22.81)	(17.97)
Percent medium-bucket credit	0.371***	0.353***	0.397***	0.354***
	(4.709)	(5.472)	(5.550)	(5.194)
Home ownership rate	0.001***	0.001***	0.001***	0.001***
	(8.853)	(6.666)	(6.850)	(4.540)
Housing Permits One Year Lag	−0.005**	−0.007***	−0.005*	−0.006***
	(−2.172)	(−3.063)	(−1.883)	(−2.703)
Unemployment Rate	0.010***	0.010***	0.012***	0.012***
	(6.420)	(5.260)	(7.325)	(5.558)
Constant	0.047	0.084**	0.096***	0.158***
	(1.555)	(2.529)	(3.138)	(4.275)
Number of observations	11,048	11,058	15,028	15,173
Adjusted *R*-squared	0.643	0.624	0.599	0.583
Cluster by state	Yes	Yes	Yes	Yes

Table 6

Consequences of credit expansion: excess demand and adverse credit outcomes.

This table presents the results of an ordinary least squares regression of changes in mortgage default and delinquency rates on measures of excess demand. Using LoanPerformance data as of December 2010, for each ZIP code in our sample we calculate the percent of loans originated in 2003, 2004, and 2005 separately that have been delinquent or in default at any point during their existence. Delinquency is defined as any loan that has been 90 days or more delinquent at any point during its existence. Default is defined as any loan that has been in the foreclosure process for at least two consecutive months. In Columns 1 and 3, the dependent variable is calculated as the difference in ZIP code–specific delinquency (default) rates between 2003 and 2004 vintage loans. Columns 2 and calculate the change in default rates between the 2003 and 2005 vintage loans. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. We discuss the construction and aggregation of the control variables to the ZIP code level in the Appendix. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Robust *t*-statistics in parentheses.

	ZIP code percent delinquent as of December 2010		ZIP code percent defaulted as of December 2010	
	2004 Vintage delinquency rate minus 2003 vintage delinquency rate (1)	2005 Vintage delinquency rate minus 2003 vintage delinquency rate (2)	2004 Vintage default rate minus 2003 vintage default rate (3)	2005 Vintage default rate minus 2003 vintage default rate (4)
Inv. bank % increase minus non-inv. Bank % Increase (2003–2004)	0.173		0.081	
	(0.884)		(0.527)	
Inv. bank % increase minus non-inv. bank % Increase (2003–2005)		1.081***		0.491*
		(3.191)		(1.965)
HPA growth lag one year	0.047	0.519***	0.057*	0.410***
	(1.034)	(5.651)	(1.940)	(6.585)
MSA avg. income—Quartile 1	4.036	12.687**	4.476**	8.346***
	(1.340)	(2.517)	(2.206)	(2.860)
MSA avg. income—Quartile 2	1.575	20.607***	1.998	13.057***
	(0.519)	(3.217)	(0.952)	(3.149)
MSA avg. income—Quartile 3	0.704	14.506**	1.508	9.055*
	(0.187)	(2.121)	(0.552)	(1.968)
MSA avg. income—Quartile 4	0.973	9.143**	0.688	4.068
	(0.355)	(2.079)	(0.353)	(1.250)
Percent low-bucket credit	3.102	1.862	1.270	−1.078
	(1.614)	(0.619)	(0.905)	(−0.454)
Percent medium-bucket credit	−15.742**	0.961	−13.314**	−2.180
	(−2.461)	(0.0884)	(−2.256)	(−0.305)
Home ownership rate	0.013	0.004	0.003	−0.008
	(1.240)	(0.167)	(0.392)	(−0.458)
Housing permits one year lag	−0.021	0.684***	0.067	0.552***
	(−0.157)	(3.377)	(0.751)	(3.312)
Unemployment rate	0.191	0.204	0.030	0.022
	(0.898)	(0.476)	(0.209)	(0.0705)
Constant	1.138	−7.874	0.882	−4.497
	(0.523)	(−1.421)	(0.633)	(−1.213)
Number of observations	11,903	11,869	11,903	11,869
Adjusted *R*-squared	0.003	0.159	0.005	0.161
Cluster by state	Yes	Yes	Yes	Yes

6.4 Tables 7–9: lax screening versus lower cost of capital

Read:

- Table 7 examines the standard deviation of interest rates within ZIP codes. The results are more consistent with reduced screening than with a pure lower-cost-of-capital mechanism.
- Table 8 shows that hard information variables such as FICO, LTV, and DTI explain more of average interest rates in positive excess-demand ZIP codes than in negative excess-demand ZIP codes.
- Table 9 shows that excess demand is positively related to the origination of no- or low-documentation loans.

Economic message: These tables support the mechanism that securitization reduced incentives to collect and price soft information. Lenders relied more on hard information and originated more low-documentation loans.

Table 7

Lax screening or lower cost of capital: evidence from the distribution of interest rates.

This table presents the results of cross-sectional regressions of changes in the variance of interest rates on measures of secondary market demand. The dependent variable is the change in the standard deviation of interest rates within a ZIP code from 2003 to 2004 and 2003 to 2005. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. Below-median FICO is an indicator equal to one if the average ZIP code FICO is below the median of the average ZIP code FICOs in the years 2004 and 2005 separately. Using our sample of securitized subprime loans, within each ZIP code we calculate the average loan interest rate, FICO score, loan-to-value (LTV) and debt-to-income (DTI) ratios, and the percentage of loans with adjustable interest rates. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Robust *t*-statistics in parentheses.

	Change in standard deviation of interest rates (2003–2004)			Change in standard deviation of interest rates (2003–2005)		
	(1)	(2)	(3)	(4)	(5)	(6)
Inv. bank % increase minus non-inv. bank % increase*below median FICO (2003–2004)			0.012			
			(1.040)			
Inv. bank % increase minus non-inv. bank % increase (2003–2004)	0.002	0.002	−0.004			
	(0.329)	(0.353)	(−0.490)			
Inv. bank % increase minus non-inv. bank % increase*below median FICO (2003–2005)						−0.002
						(−0.178)
Inv. bank % increase minus non-inv. bank % increase (2003–2005)				−0.006	−0.005	−0.004
				(−0.831)	(−0.787)	(−0.461)
Below-median FICO indicator		−0.054***	−0.060***		−0.035**	−0.033**
		(−5.372)	(−4.764)		(−2.543)	(−2.119)
Average interest rate	−0.144***	−0.149***	−0.149***	−0.275***	−0.280***	−0.280***
	(−6.012)	(−5.637)	(−5.632)	(−12.89)	(−13.75)	(−13.73)
ZIP code avg. FICO	0.001***			0.001**		
	(4.189)			(2.460)		
ZIP code avg. LTV	−0.005***	−0.005***	−0.005***	−0.004*	−0.003	−0.003
	(−2.928)	(−2.888)	(−2.915)	(−1.800)	(−1.660)	(−1.660)
ZIP code avg. DTI	0.005***	0.006***	0.006***	0.004**	0.004**	0.004**
	(3.588)	(3.715)	(3.724)	(2.649)	(2.671)	(2.671)
ZIP code percent adj. rate	0.232***	0.222***	0.224***	0.349***	0.336***	0.336***
	(3.714)	(3.564)	(3.605)	(4.481)	(4.392)	(4.414)
Constant	0.478*	1.343***	1.345***	1.724***	2.384***	2.384***
	(1.949)	(7.090)	(7.096)	(5.613)	(19.80)	(19.73)
Number of observations	15,699	15,699	15,699	15,739	15,739	15,739
Adjusted *R*-squared	0.112	0.112	0.112	0.276	0.276	0.276
Cluster by state	Yes	Yes	Yes	Yes	Yes	Yes

Table 8

Lax screening or lower cost of capital: evidence from interest rates.

This table presents the results of cross-sectional regressions of average ZIP code-level interest rates on measures of credit quality within a ZIP code. The dependent variable is the average interest rate on securitized subprime loans at the ZIP code level. We split the sample into positive and negative excess demand ZIP codes. ZIP codes in which the growth in investment bank securitization activity grew more (less) than noninvestment bank securitization activity are defined as having experienced positive (negative) excess demand. Control variables include the average FICO score, loan-to-value (LTV) and debt-to-income (DTI) ratios, and the percentage of loans with adjustable interest rates. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% levels respectively. Robust *t*-statistics in parentheses.

	Average ZIP code interest rate 2004			Average ZIP code interest rate 2005		
	Negative excess demand ZIP code (1)	Positive excess demand ZIP code (2)	Difference (3)	Negative excess demand ZIP code (4)	Positive excess demand ZIP code (5)	Difference (6)
ZIP code avg. FICO	−0.013***	−0.016***	−0.0026**	−0.016***	−0.021***	−0.0054**
	(−7.164)	(−6.498)	(−2.10)	(−8.769)	(−6.127)	(−2.54)
ZIP code avg. LTV	0.064***	0.069***	0.0049	0.067***	0.082***	0.0149**
	(9.863)	(10.06)	(1.22)	(10.40)	(9.259)	(2.83)
ZIP code avg. DTI	−0.008***	−0.012***	−0.0043	−0.010***	−0.021***	−0.0097***
	(−2.712)	(−3.179)	(−1.50)	(−3.713)	(−3.053)	(−3.71)
ZIP code percent adj. rate	−1.047***	−1.386***	−0.3389***	−1.299***	−1.768***	−0.4692***
	(−6.075)	(−7.570)	(−3.07)	(−6.495)	(−6.668)	(−3.13)
Constant	11.029***	12.460***		12.835***	15.441***	
	(11.13)	(10.83)		(15.16)	(8.959)	
Number of observations	3,492	12,292		2,666	13,191	
Adjusted *R*-squared	0.381	0.515		0.430	0.587	
Cluster by state	Yes	Yes		Yes	Yes	

Table 9

Lax screening or lower cost of capital: no or low documentation loans.

This table presents the results of cross-sectional regressions of changes in the origination of no or low documentation loans as a function of the excess demand measure of secondary market activity. The dependent variable is calculated as the change in the percent of no or low documentation loans in a given ZIP code between the years 2003–2004 and 2003–2005. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. Control variables include the average FICO score, loan-to-value (LTV) and debt-to-income (DTI) ratios, and the percentage of loans with adjustable interest rates. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Robust *t*-statistics in parentheses.

	Change in percent loans no documentation or low documentation (2003–2004)	Change in percent loans no documentation or low documentation (2003–2005)
	(1)	(2)
Inv. bank % increase minus non-inv. bank % increase (2003–2004)	0.363*	
	(1.935)	
Inv. bank % increase minus non-inv. bank % increase (2003–2005)		0.910***
		(4.860)
ZIP code avg. FICO	0.019**	0.101***
	(2.186)	(6.877)
ZIP code avg. LTV	−0.097**	−0.259***
	(−2.494)	(−6.830)
ZIP code avg. DTI	0.077**	0.148*
	(2.212)	(1.980)
ZIP code percent adj. rate	4.349***	10.060***
	(2.993)	(4.162)
Constant	−7.106	−46.471***
	(−1.024)	(−4.182)
Number of observations	15,769	15,821
Adjusted *R*-squared	0.007	0.056
Cluster by state	Yes	Yes

6.5 Tables 10–11: assignee liability event and primary-market effects

Read:

- Table 10 shows that the S&P treatment of assignee liability laws reduced investment-bank securitization activity in Massachusetts relative to other states.
- Table 11 shows that the same event-channel is associated with fewer subprime originations and higher denial rates in primary mortgage markets.
- The interaction terms validate that the excess-demand measure is sensitive to securitization-market costs.

Economic message: The event results reinforce causality: a change in securitization costs changes securitization demand and then changes primary-market credit outcomes.

6.6 Appendix Table A1: housing market elasticity robustness

Read:

- Appendix Table A1 repeats the main originations regressions with a housing market elasticity indicator as a proxy for expected house price appreciation.
- The excess-demand results remain qualitatively similar.

Table 10
The impact of the cost of securitization on excess demand.

This table estimates the impact of a change in the expected cost of securitization on measures of secondary market activity. We report the coefficients and t-statistics arising from an ordinary least squares regression of measures of secondary mortgage market activity on bank dummy variables, event dummy variables, a state dummy variable that captures a change in the expected cost of securitization, and ZIP code attributes. The dependent variable in Column 1 is calculated as the natural log of the total number of securitized loans associated with a given bank in a given ZIP code in a given year. Column 2 measures the number of securitized loans per housing unit. The dependent variable in Column 3 is calculated as the number of securitized loans associated with a given bank divided by the total number of securitized loans in a given ZIP code in a given year. The text motivates the inclusion of the event dummy variable and the Massachusetts dummy variable. Control variables include the average FICO score, loan-to-value (LTV) and debt-to-income (DTI) ratios, and the percentage of loans with adjustable interest rates. Standard errors are clustered at the state level and by year. ∗∗∗, ∗∗, and ∗ indicate significance at the 1%, 5%, and 10% level, respectively. Robust t-statistics in parentheses.

	Log of number of securitized subprime loans	Log of number of securitized subprime loans per housing unit	Bank Market Share of Securitized Subprime Loans
	sample period 2003–2006 (1)	sample period 2003–2006 (2)	sample period 2003–2006 (3)
Inv. bank∗mass. dummy∗event dummy	−0.078**	−0.248**	−0.214*
	(2.21)	(2.37)	(1.70)
Mass. dummy∗event dummy	0.121	0.077	0.059**
	(1.17)	(1.42)	(2.14)
Inv. bank dummy∗event dummy	0.262***	0.512***	0.541***
	(3.75)	(3.37)	(2.92)
Inv. bank dummy∗mass. dummy	−0.071***	−0.109	−0.097
	(3.78)	(1.26)	(1.17)
Mass. dummy	0.098	0.077	−0.198***
	(1.11)	(1.51)	(4.59)
Investment bank dummy	0.070	0.115	0.099
	(1.01)	(0.75)	(0.53)
Event dummy	−0.142***	−0.101	−0.321***
	(2.75)	(0.85)	(3.81)
ZIP code HPA t−1	0.036***	0.024***	−0.017***
	(4.30)	(9.98)	(3.49)
ZIP code avg. FICO	0.001***	−0.000	−0.001***
	(4.00)	(1.26)	(3.30)
ZIP code avg. DTI	0.004***	0.004*	0.002
	(2.68)	(1.96)	(0.82)
ZIP code avg. LTV	0.011***	0.007***	−0.005***
	(5.73)	(4.15)	(3.63)
ZIP code percent adj. rate	0.218***	0.049	−0.186***
	(4.32)	(1.20)	(5.03)
Constant	−1.091***	−7.734***	−1.475***
	(5.41)	(41.47)	(10.41)
Number of observations	907,925	679,253	907,925
Adjusted R-squared	0.102	0.076	0.084
Cluster year	Yes	Yes	Yes
Cluster by state	Yes	Yes	Yes

Table 11
Exogenous secondary market demand and primary market outcomes.

We report the coefficients and t-statistics arising from an ordinary least squares regression of measures of subprime mortgage origination activity on measures of secondary market demand and control variables. We measure access to credit in the primary mortgage market as the natural log of the number of originated subprime loans as a fraction of total housing units in a given ZIP code, or the average ZIP code mortgage denial rate. Our excess secondary market demand variable is calculated as the difference between the growth rate of secondary market activity for the investment banks subtracted by the growth rate in secondary market activity for all other securitizing banks. In this table we interact the Excess Demand variable with a dummy variable for ZIP codes from Massachusetts. We motivate the interaction term in Section 4 and discuss the control variables in the text and the appendix. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Robust t-statistics in parentheses.

	Log of higher-priced subprime loans per housing unit	Log of HUD subprime loans per housing unit	ZIP code denial rate
	2005 (1)	2005 (2)	2005 (3)
Inv. bank % increase minus non-inv. bank % increase∗Massachusetts	−0.054*	−0.008	0.006***
(2003–2005)	(−1.706)	(−0.254)	(3.224)
Inv. bank % increase minus non-inv. bank % increase	0.119***	0.126***	−0.012***
(2003–2005)	(4.246)	(4.514)	(−5.449)
Massachusetts	0.076	0.348***	−0.021***
	(1.007)	(5.361)	(-3.228)
HPA real growth lag one year	0.025***	0.037***	−0.001***
	(6.252)	(6.841)	(−4.659)
MSA avg. income—Quartile 1	0.700*	1.024**	−0.040
	(1.884)	(2.232)	(−1.165)
MSA avg. income—Quartile 2	0.680**	1.251***	−0.019
	(2.074)	(2.924)	(-0.631)
MSA avg. income—Quartile 3	0.778**	1.225***	−0.071***
	(2.635)	(3.493)	(−3.094)
MSA avg. income—Quartile 4	0.532	1.215**	−0.053*
	(1.260)	(2.131)	(−1.986)
Percent low-bucket credit	1.606***	1.961***	0.454***
	(3.567)	(4.923)	(26.69)
Percent medium-bucket credit	6.564***	5.066***	0.370***
	(7.148)	(3.392)	(5.921)
Home ownership rate	0.002	0.024***	0.001***
	(0.988)	(7.852)	(6.502)
Housing permits one year lag	0.123***	0.114***	−0.006***
	(6.555)	(5.217)	(−2.873)
Unemployment rate	0.005	0.014	0.010***
	(0.227)	(0.571)	(5.489)
Constant	−5.565***	−8.331***	0.079**
	(−12.50)	(−15.36)	(2.329)
Number of observations	12,182	10,442	12,183
Adjusted R-squared	0.303	0.290	0.632
Cluster by state	Yes	Yes	Yes

Economic message: The robustness check reduces concern that the main results simply reflect expected house-price appreciation in elastic or inelastic housing markets.

Table A1
Secondary market demand and access to credit: housing market elasticity and expected rates of house price appreciation.

We estimate cross-sectional regressions on data from the years 2004 and 2005 separately. The regressions measure the impact of demand from the secondary mortgage market on the extension of credit in the primary mortgage market. Access to credit in the primary mortgage market is measured as the natural log of the number of originated subprime loans as a fraction of total housing units in a given ZIP code. Subprime loan originations are measured using the Home Mortgage Disclosure Act (HMDA) higher-priced definition and the Department of Housing and Urban Development (HUD) subprime lender list. See Appendix for details. Our excess demand measure of secondary market demand is calculated as the difference between growth in the securitization activity of the broker/dealer investment banks subtracted by the growth rate in securitization activity for all other securitizing banks. Inelastic Housing Market Indicator is a dummy variable equal to one for ZIP codes with a below-median housing market elasticity measure (Saiz, 2009). We discuss the construction and aggregation of the control variables to the ZIP code level in the data appendix. Standard errors are clustered at the state level. ***, **, and * indicate significance at the 1%, 5%, and 10% level, respectively. Robust t-statistics in parentheses.

	(1)	(2)	(3)	(4)
	Log of higher-priced subprime loans per housing unit		Log of HUD subprime loans per housing unit	
Inv. bank % increase minus non-inv. bank % increase	0.038***		0.016	
(2003–2004)	(2.782)		(0.505)	
Inv. bank % increase minus non-inv. bank % increase		0.067***		0.058**
(2003–2005)		(3.639)		(2.398)
Below-median housing market elasticity indicator	0.188***	0.327***	0.269***	0.372***
	(3.245)	(4.635)	(2.987)	(4.041)
MSA avg. income—Quartile 1	0.470	0.876**	1.601***	1.883***
	(1.505)	(2.625)	(3.622)	(4.370)
MSA avg. income—Quartile 2	0.677*	1.196**	2.272***	2.513***
	(1.679)	(2.408)	(3.318)	(3.740)
MSA avg. income—Quartile 3	0.476	1.079**	1.920***	2.219***
	(1.316)	(2.521)	(4.468)	(4.267)
MSA avg. income—Quartile 4	0.314	0.653*	1.712***	1.777***
	(0.884)	(1.689)	(3.634)	(3.787)
Percent low-bucket credit	2.079***	1.430***	1.136***	1.530***
	(6.332)	(3.888)	(3.324)	(4.383)
Percent medium-bucket credit	6.629***	6.735***	6.726***	6.246***
	(7.299)	(6.157)	(4.976)	(3.776)
Home ownership rate	0.003	−0.000	0.020***	0.019***
	(1.622)	(−0.181)	(8.246)	(6.150)
Housing permits one year lag	0.129***	0.159***	0.148***	0.162***
	(6.088)	(8.431)	(5.850)	(7.529)
Unemployment rate	−0.003	−0.015	0.025	−0.011
	(−0.109)	(−0.428)	(0.636)	(−0.233)
Constant	−5.834***	−5.298***	−8.291***	−8.283***
	(−11.83)	(−9.693)	(−13.37)	(−11.32)
Number of observations	11,101	11,120	9,836	9,032
Adjusted R-squared	0.299	0.274	0.239	0.246
Cluster by state	Yes	Yes	Yes	Yes

7 Short summary

- The paper studies whether securitization caused the expansion of subprime credit in 2003–2005.
- It constructs an Excess Demand measure based on the growth of investment-bank securitization relative to noninvestment-bank securitization within ZIP codes.
- The treated banks are Bear Stearns, Lehman Brothers, Merrill Lynch, Morgan Stanley, and Goldman Sachs.

- Higher excess demand is associated with more subprime originations and lower denial rates.
- Higher excess demand is also associated with higher subsequent delinquency and default rates.
- The mechanism evidence favors weaker screening incentives over a pure funding-cost story.
- Interest-rate dispersion falls, hard information explains more pricing, and low/no documentation lending rises in high excess-demand areas.
- A 2004 S&P treatment of assignee liability laws provides an external securitization-cost event that supports the secondary-market interpretation.
- The paper shows that private-label securitization actively shaped mortgage supply rather than passively absorbing loans.

8 Conclusion

8.1 Core conclusion

Nadauld and Sherlund show that the securitization boom was a causal force behind the expansion of subprime mortgage credit. ZIP codes exposed to greater excess demand from broker/dealer investment banks experienced more subprime lending, lower denial rates, and worse ex post credit outcomes.

8.2 Economic intuition

The intuition is that securitization affects originators' incentives. When investment banks demand more securitized collateral, originators can sell more loans into the secondary market. This expands credit supply and weakens screening because the originator does not bear the full consequences of poor loan performance. The result is more lending but lower average loan quality.

8.3 Incremental contribution

The paper's incremental contribution is to separate secondary-market securitization demand from primary-market borrower demand using matched loan-deal data and within-ZIP variation across securitizing-bank types. This allows the authors to show that securitization was not merely correlated with the subprime boom; it helped cause the boom and shaped the quality of loans originated during it.

8.4 Takeaway

The study provides evidence that securitization-market structure can alter real credit allocation. Private-label securitization can expand household credit, but if it weakens screening incentives, the expansion may generate higher defaults and financial fragility.